

Design FMEA (qualitative version) — V11 (t1_r3)

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Note: qualitative version based on simulation risk signals; severity (S) and occurrence (O) are three-level qualitative ratings (high/medium/low) whose basis is the magnitude of the simulation value relative to its threshold. Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary).

#	Failure mode	Failure cause	Simulation signal (detectability basis)	S	O	Design-side mitigation
1	Lithium plating on fast charge	local anode potential < 0 V during 4C	`r5_v11_4c.json:anode_potential_v` min = +0.0425 V (42.5 mV margin at true 4C / 20.34 A)	high	low	0.8 µm particles (2× surface), electrolyte D 6.0e-10 m ² /s / t _{max} 0.55, 4C current capped at 20.34 A; note: cold-ambient fast charge not simulated — do not 4C-charge below 0 °C without physical validation
2	Thermal runaway / overheat	excessive heat generation vs cooling	4C T _{max} 321.2 K vs 333.15 K limit (margin 12.0 K); overcharge 4.7 V T _{max} 298.57 K; TR ODE triggered = false	high	low	h = 120 W/m ² /K cell cooling design; abuse tolerance verified by coupled overcharge→TR simulation
3	Electrolyte oxidative decomposition at high voltage	electrolyte HOMO above positive potential window	cell-scale: no thermal signal in 4.7 V overcharge (`r5_v11_oc.json:T_max_K` = 298.57 K), but DFN has no oxidation kinetics and molecular-level HOMO endorsement skipped (real_compute=false)	medium	medium	operate at ≤ 4.2 V; 4.7 V only as abuse-tolerance margin; recommend true DFT HOMO/IE-EA endorsement before production
4	Insufficient capacity / energy	capacity below task target	1C capacity 5.0849 Ah; ED 605.33 Wh/kg vs 392.61 (54% margin)	low	low	ED margin maintained by thin collectors (10/6 µm) and thin separator (8 µm)
5	Mechanical integrity of thinned components	6 µm Cu foil / 8 µm separator / 10 µm Al foil handling and dendrite puncture	no simulation signal (purely mechanical domain) — qualitative	medium	medium	manufacturing QC on foil handling; separator puncture/dendrite risk bounded by plating-free margin (+0.0425 V); physical mechanical tests N/A in virtual design

Conclusion

Highest-risk items: #1 and #2 (safety-critical) — both mitigated in-design with verified simulation margins (plating +42.5 mV, temperature 12 K below limit, no TR trigger after 4.7 V overcharge). Item #3 remains an open molecular-level question (honestly flagged; real_compute=false). Items #4/#5 low-to-medium.